

# India and Global Rare-Earth Permanent Magnet (REPM) Demand: End-Market Quantification and 2030/2035 Outlook

*Institutional research note — sintered/bonded NdFeB (plus SmCo where relevant). All primary demand figures 2025/2026 vintage; pre-2025 items flagged. Current date: 15 August 2026.*

## TL;DR

- India consumes c.4,000 tonnes of finished REPM a year (Government of India/IREL estimate), essentially 100% imported; the genuine NdFeB slice from China was c.2,850 tonnes in 2024 (Adamas). The widely-quoted “53,748 tonnes FY25 import” is gross HS-8505 tonnage that is c.93% Chinese-sourced but overwhelmingly cheap ferrite — do not read it as NdFeB. Official demand roughly doubles to 8,220 tonnes by 2030 (EV 3,250 t, wind 1,800 t, BLDC fans 980 t). Newsage
- Global finished NdFeB demand is c.240,000–260,000 tonnes (2023–24; Adamas/SMM), c.94% made in China (IEA, 2024 figure); automotive is the largest end-market at c.33–42%, consumer electronics c.20–24%, wind c.15–19%, defence c.1–2%. NdFeB is 96% of the rare-earth magnet market (IDTechEx, Jul 2025).
- The binding constraint is NdPr/heavy-rare-earth feedstock, not magnet-pressing. Adamas forecasts a 60,000 t/yr NdFeB alloy/powder deficit by 2030. India’s 6,000 MTPA scheme can in principle cover most 2030 domestic demand, but the government’s own 5,000 t-by-2030 production trajectory implies continued imports. Robotics/humanoids are the real wildcard — highly speculative, but potentially the single largest driver by 2040.

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## Key Findings

**1. The Indian import number is routinely misquoted.** Three different quantities circulate in the ₹-scheme press cycle and must be kept distinct: (a) **53,700–53,748 tonnes** of ALL permanent magnets imported in FY25 (gross customs/DGCI, c.93% from China, mostly ferrite); (b) **c.2,850 tonnes** of NdFeB magnets + alloy from China in CY2024 (Adamas — the number that matters for this note); and (c) **c.4,010 tonnes** of finished REPM annual

demand (Gol / IREL–DMRL–BARC assessment). Per Business Standard/IndraStra (Jun 2025), the FY25 import value grew only c.5% to **₹1,744 crore** even as tonnage nearly doubled [IndraStra Global](#) — and SBI Research pegs magnet imports at **\$291 million** in FY25. [Sbi](#) On 53,748 t that is a blended c.\$5.4/kg, versus \$50–100/kg for sintered NdFeB — arithmetic proof the tonnage is dominated by low-value ferrite, not the rare-earth magnets this note is about.

**2. India’s end-market mix is EV- and appliance-heavy, and uniquely 2W/3W-weighted.**

The only *official* India end-market split is the 2030 forecast (EV 3,250 t / wind 1,800 t / BLDC fans 980 t of 8,220 t total). India’s e-2W volumes (1.4 million units in FY26, 57–58% of all EV sales) dominate EV units but consume modest NdFeB per vehicle, so 2W/3W matter more in unit terms than tonnage terms.

**3. China dominance is near-total and the April 2025 export-control regime bit India hard.** China makes 94% of sintered magnets (IEA, 2024 figure; refining share 91%). [IEA](#)

The **4 April 2025** MOFCOM controls on seven medium/heavy rare earths plus magnets [International Energy Agency](#) forced Indian and global automakers to cut output; [datadeep](#) conditional licences to Indian firms resumed from October 2025; [Electrive](#) a broader **October 2025** regime (five more elements, overseas products made with Chinese inputs) was **suspended until November 2026** after the Xi–Trump Busan meeting, [China Briefing](#) but the April controls remain in force and heavy-REE/SmCo/Dy-Tb shipments stay slow and case-by-case through 2026 (S&P Global, Arnold Magnetics).

**4. Global demand estimates diverge because of definitions, not disagreement.** The “385,000 t” 2025 figure (Future Markets Inc / Research and Markets) is a c.50% outlier versus the primary-source consensus (240k–260k t finished magnets). The gap is magnet-tonnage vs alloy vs NdPr-oxide vs revenue, plus inclusion/exclusion of captive Chinese consumption.

**5. The deficit is real and structural.** Adamas: 60,000 t/yr NdFeB alloy/powder undersupply by 2030 and 246,000 t/yr by 2040; [Adamas Intelligence](#) NdPr-oxide undersupply c.19,000–21,000 t by 2030. IEA: ex-China magnet-rare-earth demand rises c.one-third by 2030, but operating + planned ex-China projects meet only c.a quarter of refining demand and **less than one-fifth of magnet demand by 2035.** [Aerospace-and-defence](#)

Details

1. India — Current-Year REPM Consumption (reconciliation table)

|  |  |  |  |         |
|--|--|--|--|---------|
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|--|--|--|--|---------|

| Metric                                   | Figure                                                   | Source                                                  | Date           | measures                                               |
|------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|----------------|--------------------------------------------------------|
| Total permanent-magnet imports, FY25     | 53,748 t (c.93% China)                                   | Govt data via Business Standard / electrive / IndraStra | Jun & Dec 2025 | ALL HS-8505 magnets, mostly ferrite — <b>NOT NdFeB</b> |
| Same, FY24                               | 28,700 t                                                 | electrive                                               | Dec 2025       | Gross, prior year                                      |
| Magnet import value, FY25                | ₹1,744 cr / \$291 m                                      | Business Standard; SBI Research                         | Jun 2025       | All magnets (4-yr avg \$249 m)                         |
| NdFeB magnets + alloy from China, CY2024 | c.2,850 t (+49% YoY; India = 5th-largest China importer) | Adamas Intelligence                                     | Mar 2025       | The relevant NdFeB number                              |
| Finished REPM annual demand ("current")  | c.4,010 t                                                | Goi / IREL–DMRL–BARC                                    | 2025–26        | Finished REPM consumption                              |
| Older/narrower press estimate            | c.900 t                                                  | Sanskriti IAS / press                                   | 2025           | Dated/narrow                                           |

**PUBLISHED FACT:** India imports essentially 100% of finished REPM; commercial NdFeB production is nil. **INFERENCE:** the defensible current-consumption anchor is **c.4,000 t of finished REPM**, of which the NdFeB portion is c.2,850–3,500 t.

**India current-year end-market breakdown (ESTIMATE/INFERENCE — no official current-year split is published):**

| End market                                               | Est. tonnes   | Est. %   | Basis                                                                             |
|----------------------------------------------------------|---------------|----------|-----------------------------------------------------------------------------------|
| Automotive total (EV traction + ICE ancillaries + 2W/3W) | c.1,500–1,700 | c.38–42% | Scaled from official 2030 EV share; bottom-up on e-2W/3W                          |
| — EV traction (4W+2W+3W)                                 | c.900–1,100   | —        | 1.4 m e-2W (FY26) × 0.1–0.3 kg; e-cars (c.194k, FY26) × c.1.5 kg; e-3W × c.0.3 kg |
| — ICE ancillaries (EPS, sensors, small motors)           | c.500–700     | —        | c.0.3–0.6 kg/ICE vehicle                                                          |
|                                                          |               |          |                                                                                   |

|                                                     |             |          |                                                                   |
|-----------------------------------------------------|-------------|----------|-------------------------------------------------------------------|
| Consumer electronics & appliances (incl. BLDC fans) | c.900–1,100 | c.24–27% | Official 980 t BLDC-fan demand by 2030 implies large current base |
| Wind energy                                         | c.400–600   | c.12–15% | Direct-drive share of India installs                              |
| Industrial motors, pumps, HVAC compressors          | c.400–500   | c.10–12% | IE3/IE4 efficiency push                                           |
| Defence, aerospace, space, medical (MRI), other     | c.150–250   | c.4–6%   | Small tonnage, strategically critical, poorly disclosed           |

**DATA GAP (flag):** India publishes no current-year end-market tonnage split. The above is a bottom-up researcher estimate anchored to the official 2030 shares and Vahan/SIAM volumes — indicative, not precise.

**India's official 2030 demand split (PUBLISHED FACT — GoI written reply to Lok Sabha, Jul 2026, IREL–DMRL–BARC assessment):** Total **8,220 t**; EV **3,250 t** (largest); wind **1,800 t**; BLDC fans **980 t**; remainder smartphones, computers, industrial motors, other.

Newsage

## 2. India — Domestic Production and the ₹7,280-crore Scheme

- **PUBLISHED FACT:** Commercial NdFeB output is nil. IREL runs a SmCo REPM plant at Visakhapatnam (operational 2023) and produced c.40 tonnes of neodymium last year, targeting a multi-fold rise to c.200 tonnes by FY27-end [Business Standard](#) (IREL, Nov 2025). Midwest Advanced Materials targeted commercial NdFeB for EV motors from late 2025. [FinancialContent](#)
- **The scheme (PUBLISHED FACT — PMIndia release, 26 Nov 2025; notified 15 Dec 2025):** Outlay **₹7,280 crore** (c.\$802–816 m) = **₹6,450 cr** sales-linked incentives (5 yr) + **₹750 cr** capital subsidy. [Prime Minister of India](#) Target **6,000 MTPA** integrated sintered NdFeB across five beneficiaries (600–1,200 MTPA each, in multiples of 100). [Outlook Business](#) Seven-year duration (2-yr gestation + 5-yr incentive). [Prime Minister of India](#) **IREL to supply assured NdPr oxide to the three lowest bidders.** [Outlook Business](#) RFP issued 20 March 2026; [Outlook Business](#) **20 bids received** (Business Standard, 13 Aug 2026). [Business Standard](#) Vedanta (Anil Agarwal) confirmed intent to bid; [Swarajya](#) Sona Comstar and Uno Minda expressed interest.
- **Government production trajectory (PUBLISHED FACT — Jitendra Singh, Lok Sabha, Mar 2026):** 500 tpa phase 1 → 2,000 t by 2028 → 5,000 t by 2030, [ETV Bharat](#) against demand rising to c.8,000 t. Minister Kishan Reddy (Reuters, 19 Feb 2026, at FICCI) said India “aims to start producing rare earth permanent magnets by the end of the year.”

[Yahoo Finance](#) India holds 6.9 million tonnes of REE reserves (USGS, world's third-largest) but mines a fraction. [Yahoo Finance](#)

- **INFERENCE:** "20 bids ≠ 6,000 t of qualified magnets." The genuine risks are NdPr/Dy/Tb feedstock security, oxide-to-metal conversion, and automotive/defence qualification — precisely where China's advantage is deepest. Finished REPMs are c.43% costlier in India than abroad [Business Standard](#) (GoI documents), which is why the sales-linked incentive exists.

### 3. World — Current REPM Consumption

**Global totals (reconciled):**

| Metric                                                      | Figure                                     | Source                   | Date     |
|-------------------------------------------------------------|--------------------------------------------|--------------------------|----------|
| Global finished RE-PM production (NdFeB + minor SmCo), 2023 | c.240,000 t                                | Adamas Intelligence      | 2024     |
| China RE-PM production 2023 (+14% YoY); 2024e               | c.240,000 t; c.260,000 t                   | SMM                      | 2024     |
| Global neodymium magnet production, 2024                    | 220,000–240,000 t (≥85%, poss. 90%, China) | Grand View Research      | 2025     |
| China share of sintered magnets, 2024                       | 94% (refining 91%)                         | IEA                      | Oct 2025 |
| China RE-PM + alloy exports, 2024                           | 58,152 t (\$2.86 bn)                       | China GAC via Adamas     | Mar 2025 |
| NdFeB share of RE-magnet demand, 2025                       | 96%                                        | IDTechEx                 | Jul 2025 |
| Outlier (flag)                                              | c.385,000 t (2025)                         | Future Markets Inc / R&M | Oct 2025 |

**Global end-market breakdown (2024–25; directional consensus — do NOT sum across differing-definition sources):**

| End market       | Share/tonnage               | Source                  |
|------------------|-----------------------------|-------------------------|
| Automotive total | c.33% (2024) → 41.8% (2025) | Grand View; SNS Insider |

|                              |                                                                           |                |
|------------------------------|---------------------------------------------------------------------------|----------------|
| — EV traction only           | c.38,000 t (2025)                                                         | Dataintel      |
| Consumer electronics         | c.20–24% (23.6% in 2026)                                                  | Fact.MR        |
| Wind                         | c.15–19% globally (higher regionally; up to 60% in IEA net-zero scenario) | consensus; IEA |
| Industrial motors / robotics | fast-growing; robotics = largest driver by 2040                           | Adamas         |
| Defence / aerospace          | c.1–2%                                                                    | —              |
| — US military consumption    | 3,000–4,000 t/yr (potentially tripling by 2030)                           | CSIS, 2025     |

**China geographic supply (2024):** Germany was the largest China importer (c.11,000 t, +24%), US second (c.7,500 t), South Korea third (c.5,800 t), Vietnam fourth (c.4,750 t), India fifth (c.2,850 t) [adamasintel](#) — Adamas. [Adamas Intelligence](#)

#### 4. Magnet Intensity — the Swing Factor in Every Forecast

| Application                     | NdFeB intensity                                                  | Source                           |
|---------------------------------|------------------------------------------------------------------|----------------------------------|
| BEV traction motor              | 1–2 kg (avg c.1.5 kg; 0–4 kg range)                              | Thunder Said Energy; Persistence |
| — per vehicle, NdPr-oxide basis | c.0.7 kg (127 kW sales-weighted avg motor, 2025)                 | Adamas                           |
| Hybrid / EV avg                 | c.1.25 kg                                                        | academic (arXiv)                 |
| ICE ancillaries (ex-engine)     | c.0.6 kg                                                         | MDPI (2023, flag vintage)        |
| e-2W (India)                    | c.0.1–0.3 kg                                                     | researcher estimate              |
| Direct-drive wind (PMSG)        | 500–650 kg/MW (160–1,000 range); c.165 kg/MW REE content         | IRENA; ScienceDirect             |
| DFIG/geared wind                | far lower (declining direct-drive penetration per SMM, Jun 2025) | SMM                              |
|                                 |                                                                  |                                  |

|                |                                   |                          |
|----------------|-----------------------------------|--------------------------|
| Humanoid robot | 30–50 small motors; c.1.3 kg NdPr | IDTechEx; Morgan Stanley |
|----------------|-----------------------------------|--------------------------|

**INFERENCE:** Because EV traction uses c.1.5 kg while a direct-drive turbine uses c.500 kg/MW, EV's share of *finished-magnet tonnage* (c.33–42%) is far larger than its share of *magnet rare-earth demand* (IEA: c.9% now → 18% by 2030). [Saur Energy International](#)  
Analysts must specify which base they mean.

## 5. Projections — 2030 and 2035 (range across sources)

### Global demand growth (PUBLISHED FORECASTS):

- **Adamas:** NdFeB magnet demand +7.5% CAGR 2023→2040 (c.4× expansion), [MiningNewsWire](#) with didymium/Dy/Tb supply rising at a slower 5.2% CAGR; [Adamas Intelligence](#) 9% CAGR through 2035 (2025 conference). [Adamas Intelligence](#)
- **IEA:** magnet-rare-earth demand >120 kt by 2030 and 175 kt by 2050; [Saur Energy International](#) ex-China demand +50% by 2035.
- **IDTechEx:** rare-earth magnet demand 332 ktpa by 2036. [IDTechEx](#)

**Deficits (Adamas, PUBLISHED):** NdFeB alloy/powder undersupply **60,000 t/yr by 2030 and 246,000 t/yr by 2040** (Outlook to 2040 vintage) — “almost equal to last year’s total global NdFeB alloy and powder production.” [Adamas Intelligence](#) Earlier vintage: 66,000 t (2030) / 206,000 t (2035). [Adamas Intelligence](#) NdPr-oxide undersupply c.19,000–21,000 t by 2030, 68,000 t by 2035. Adamas: US + Europe demand to exceed combined capacity by **65,000 t by 2035**. [Adamas Intelligence](#) IEA: ex-China projects meet <20% of 2035 magnet demand.

### Bull / Base / Bear framework (researcher synthesis of the above):

| Scenario    | Global 2035 NdFeB demand                             | Key drivers                                                                                                 |
|-------------|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <b>Bull</b> | >400,000 t; acute deficit                            | Fast EV S-curve + wind buildout + humanoid inflection by 2030; minimal substitution                         |
| <b>Base</b> | c.350,000 t (IDTechEx 332 ktpa 2036; Adamas 9% CAGR) | Central EV/wind; deficit c.60,000 t (2030) partly filled by ex-China ramp + recycling                       |
|             |                                                      | Rare-earth-free motors (BMW/Renault/ZF externally-excited synchronous), ferrite/induction, aggressive Dy/Tb |

|      |                   |                                                           |
|------|-------------------|-----------------------------------------------------------|
| Bear | Materially softer | thriftig, recycling reaching c.10% of supply by mid-2030s |
|------|-------------------|-----------------------------------------------------------|

**The humanoid wildcard (flag as SPECULATIVE):** Estimates range from 3,400 t (at 1 m units) to 34,000 t of incremental NdFeB (couragemagnet); Couragemagnet Adamas says robotics surpasses all other end-uses by 2040; Value The Markets Elon Musk's 10-billion-robot figure would need >100x current global NdFeB output (Katusa, citing Adamas).

Katusa Research These are scenario projections, not demand-in-hand — Unitree shipped only c.5,500 humanoid units in 2025. Rare Earth Exchanges

**India 2030/2035:** Official demand 8,220 t by 2030. Scheme capacity of 6,000 MTPA could cover c.70%+ of 2030 demand *if fully commissioned and qualified*, but the government's own 5,000 t-by-2030 production path implies continued imports of c.3,000+ t. India 2035 demand is not officially published; **researcher estimate c.12,000–16,000 t** on continued EV/wind/appliance growth, of which domestic supply might cover half at best.

## 6. Ex-China Capacity Buildout

- **Total non-Chinese NdFeB capacity c.8,000–10,000 t/yr** (ForcedAlpha): Japan c.4,500 t (TDK, Shin-Etsu, Proterial); Europe c.2,000 t (Neo Performance's Narva, Estonia plant, live since Sept 2025, phase 2 to 5,000 t); US 1,600–2,000 t commissioning (MP Materials, USA Rare Earth, Noveon).
- **MP Materials (Independence, Fort Worth TX):** c.1,000 t/yr NdFeB; FY2025 record 2,599 t NdPr oxide (+101% YoY), datadeep Magnetics revenue \$66.9 m (first substantial year), ceased all China sales July 2025 (10-K).
- **Vulcan Elements:** 10,000 t/yr target, \$1.4 bn public-private funding (\$620 m DoD loan). e-VAC Magnetics (Sumter, SC): supplying GM, expandable to 12,000 t. Neo Estonia: 2,000 t → 5,000 t.
- **JL MAG (China):** c.40,000 t capacity end-2025 (34,400 t magnetic materials produced in 2025, net profit +142% to CNY706 m), targeting 60,000 t by 2027 — a single Chinese producer's expansion (+20,000 t) roughly matches all announced Western greenfield combined, underscoring the scale asymmetry.

## Recommendations

1. **Model India's investable NdFeB demand off c.3,000–4,000 t today, not the 53,748 t headline.** For auto-OEM/component coverage, the near-term tonnage swing is EV traction + EPS/sensor content; use 1.5 kg/BEV, 0.3–0.6 kg/ICE and 0.1–0.3 kg/e-2W. The

53,748 t figure is ferrite-dominated and will mislead any demand model that treats it as rare-earth magnets.

2. **Treat the 6,000 MTPA scheme as a 2028–2031 story, not 2026.** Benchmarks that would move the thesis: (a) beneficiary awards post the 20-bid RFP; (b) IREL NdPr oxide reaching c.200 t (FY27) then scaling; (c) first *automotive-qualified* domestic sintered NdFeB shipment. Slippage on NdPr feedstock is the trigger to cut domestic-supply assumptions and extend the import-dependence horizon.
  3. **Position for structural NdPr tightness through 2030, but stress-test the bear case.** If rare-earth-free motors and recycling scale faster than expected, the 60,000 t (2030) deficit narrows — monitor OEM motor-architecture announcements (BMW/Renault/ZF EESM adoption) as the leading indicator.
  4. **Flag humanoids as optionality, not base case.** Do not embed large humanoid tonnage in a base-case model before unit economics and volumes are proven; size it as a call option on the 2035+ horizon.
  5. **Watch the April-2025-vs-October-2025 control distinction.** The October regime's suspension to Nov 2026 is a reversible truce; the April controls (seven REEs + magnets) remain live. Renewed tightening is the single biggest catalyst for the ex-China capacity trade and for Indian localisation urgency.
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## Caveats

- India's current-year end-market split is a researcher estimate; only the 2030 forecast (8,220 t) is official (IREL–DMRL–BARC).
- Global end-market percentages come from sources using different definitions (finished-magnet tonnage vs magnet-rare-earth value vs revenue) and must not be summed.
- Defence tonnage is poorly disclosed for both India and globally; the 3,000–4,000 t/yr figure is US-only (CSIS).
- The 385,000 t global 2025 figure is a flagged outlier c.50% above the primary-source consensus; the finished-magnet consensus is 240k–260k t (2023–24). A clean, named 2025 global finished-magnet tonnage sits behind Adamas/IDTechEx paywalls.
- Magnet tonnage ≠ magnet-alloy tonnage ≠ NdPr-oxide tonnage ≠ total REO tonnage; conversion factors differ materially and are the primary reason cross-source figures appear to conflict.
- One intensity datapoint (MDPI ICE-ancillary 0.6 kg) is 2023 vintage, flagged accordingly.

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## Source Appendix (with publication date and use)

- **PMIndia release** (26 Nov 2025) — ₹7,280 cr scheme structure, 6,000 MTPA, five beneficiaries.
- **Business Standard / IndraStra** (Jun 2025; 13 Aug 2026) — FY25 imports 53,748 t (93% China), ₹1,744 cr value, 20 bids received, 8,220 t/2030 demand.
- **electrive.com** (12 Dec 2025) — FY25 53,748 t vs FY24 28,700 t; China licence resumption Oct 2025.
- **SBI Research via Deccan Herald** (Jun 2025) — \$291 m magnet imports FY25, five impacted sectors.
- **Adamas Intelligence** — “Record RE magnet exports from China 2024” (Mar 2025): 58,152 t China exports, India c.2,850 t; “Outlook to 2040” (2024/Dec 2025): 240k baseline, 7.5% CAGR, 60,000 t (2030)/246,000 t (2040) deficit; humanoid & robotics insights (2025–26).
- **IEA** — “Rare Earth Elements” report & commentary (2025/Oct 2025): 94% China sintered-magnet share, April/October 2025 controls, ex-China <20% of 2035 magnet demand, EV 9%→18%, >120 kt by 2030.
- **IDTechEx** — “Rare Earth Magnets 2026-2036” (Jul 2025): NdFeB 96% of 2025 demand, 332 ktpa by 2036.
- **SMM (Shanghai Metals Market)** (2024–25) — China 240k (2023)/260k (2024e) production; DFIG penetration rising.
- **Grand View Research** (2025) — 220k–240k t 2024 neodymium magnet production; auto >33% (2024).
- **SNS Insider** (Jul 2026) — automotive 41.78% of NdFeB, 2025.
- **Fact.MR** (2026) — consumer electronics 23.6%, 2026.
- **Dataintelo** (2025–26) — EV traction c.38,000 t 2025.
- **CSIS** (2025) — US military 3,000–4,000 t/yr.
- **Gol / Lok Sabha replies** (Mar & Jul 2026) — 4,010 t current demand, 8,220 t/2030 split (EV 3,250/wind 1,800/BLDC fans 980); 500→2,000→5,000 t production path.
- **IREL via Business Standard** (Nov 2025) — 40 t Nd (last yr) → c.200 t by FY27.
- **Reuters** (19 Feb 2026) — Kishan Reddy at FICCI, “start producing by year-end”; 6.9 m t

reserves (USGS).

- **Vahan / Autocar Professional / EVreporter / JMK Research** (2025–26) — India EV volumes: 1.4 m e-2W FY26, 2.55 m total EVs, 8.5–8.64% penetration.
- **MP Materials 10-K** (FY2025, SEC) — 2,599 t NdPr oxide (+101%), c.1,000 t NdFeB, \$66.9 m Magnetism revenue, China sales ceased Jul 2025.
- **Neo Performance / Fastmarkets / Discovery Alert** (2025) — Narva Estonia 2,000→5,000 t, live Sept 2025.
- **Vulcan Elements / Discovery Alert** (2025) — 10,000 t target, \$1.4 bn funding.
- **JL MAG annual report / Yicai / filingreader** (2025–26) — 40,000 t capacity, 34,400 t output 2025, 60,000 t by 2027.
- **ForcedAlpha** (2025) — non-China capacity 8,000–10,000 t/yr; Japan c.4,500 t.
- **Thunder Said Energy; Persistence; ScienceDirect; arXiv; MDPI (2023, flagged)** — magnet intensity ranges.
- **Arnold Magnetism; S&P Global** (2025–26) — 2026 supply-chain disruption outlook.
- **China Briefing; Clark Hill; European Parliament** (2025) — October 2025 controls suspended to Nov 2026; April controls remain.
- **Katusa Research; couragemagnet; RareEarthExchanges** (2025–26) — humanoid demand scenarios (speculative).